

SADP

Course: BTECH (CS01)

Set: A | Time: 3 Hours | Max Marks: 35 | Date: 15-03-2026

Name: _____

Roll No: _____

Student Sign: _____

Teacher Sign: _____

GENERAL INSTRUCTIONS:
Write all the questions

SECTION A: MCQs

1. Which of the following is a core characteristic of an embedded system? **[2 Marks]**

- a) It is designed for general-purpose computing tasks.
- b) It performs a specific, dedicated function within a larger system.
- c) It requires a large amount of user-replaceable hardware.
- d) It always uses a high-performance graphical user interface.

2. In an embedded system, what is the primary difference between a microcontroller and a microprocessor? **[2 Marks]**

- a) A microcontroller lacks an internal CPU.
- b) A microprocessor includes integrated RAM, ROM, and I/O on a single chip.
- c) A microcontroller integrates the CPU, memory, and peripherals on a single chip.
- d) Microprocessors are only used for real-time applications.

3. Which communication protocol uses only two wires, SDA and SCL, to connect multiple integrated circuits? **[2 Marks]**

- a) SPI
- b) UART
- c) I2C
- d) CAN

4. Which mechanism is used in Real-Time Operating Systems (RTOS) to solve the problem of priority inversion? **[2 Marks]**

- a) Context Switching
- b) Priority Inheritance Protocol
- c) Round Robin Scheduling
- d) FIFO Buffer Management

5. What type of memory is typically used to store the firmware or boot code in an embedded system? **[2 Marks]**
- a) RAM
 - b) ROM or Flash Memory
 - c) Cache Memory
 - d) Virtual Memory

SECTION B: SHORT ANSWERS

6. Define an embedded system and provide two common examples. **[5 Marks]**
7. Explain the purpose and operation of a Watchdog Timer in embedded hardware. **[5 Marks]**
8. What is interrupt latency and why is it a critical factor in embedded system design? **[5 Marks]**
9. Differentiate between Hard Real-Time and Soft Real-Time systems. **[5 Marks]**
10. Describe the Harvard Architecture and its advantage in embedded processors. **[5 Marks]**

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